# Post-Game Report: Catcher Evaluation

game_id 01103572  catcher_id zkrGGa8L

## 1. Evaluation Metrics

For the evaluation, the following stats will be used:

- Challenge-Worthy Called Balls Seen (A called ball is **challenge-worthy** if our model predicts the challenge success probability to be over 80%)
- Good Challenge Rate(+) (explained below)
- Challenge Success Rate (% challenge succeeded)
- Calls Overturned (number of successful challenges)
- Average Unsuccessful Challenge Miss Distance (how much the challenged balls were off the zone by in inches)

**GCR (Good Challenge Rate)**: *Measures the percentage of **challenge-worthy** called balls the catcher challenged.*

$$GCR = \frac{\text{\# of } \textbf{challenge-worthy} \text{ calls challenged}}{\text{\# of } \textbf{challenge-worthy} \text{ called balls seen*}}$$

**called balls seen **while** having challenges available. A $GCR$ of 0.300 means the catcher capitalized on 30% of promising $ABS$ challenges.*

**GCR+**: *The plus-stat version of $GCR$.*

$$GCR+ = \frac{GCR}{\text{league average } GCR} \times 100$$

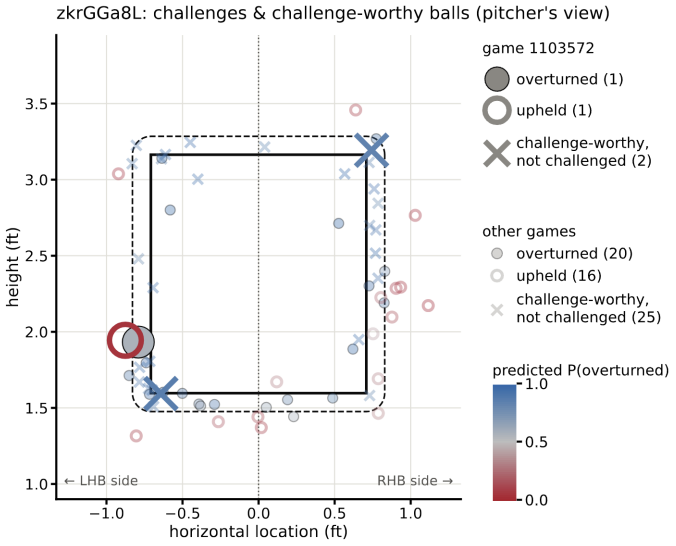
*100 is league average performance. A catcher with a $GCR+$ of 120 means he is 20% better than the league average catcher with regards to capitalizing on promising ABS challenges opportunities.*

## 2. Performance vs. League Average

*The table and figure below show the distribution of **single game stats**. Note that single game stats have a different (usually noisier and more discrete) distribution compared to **accumulated stats** calculated across a window or a season. Since we are evaluating a single game's performance, we will be comparing against single game stats.*

| | This game | Season | Single-Game League Median | Season League Avg. |
|---|---|---|---|---|
| CWCBS | 2 | 1.538 | 1 | 1.591 |
| $GCR$ | 0 | 0.375 | 0 | 0.300 |
| $GCR+$ | 0 | 125 | 0 | 100 |
| Challenge Success Rate | 0.500 | 0.556 | 0.500 | 0.532 |
| Calls Overturned | 1 | 0.951 | 0 | 0.545 |
| AUCMD | 0.559 | 1.102 | 0.898 | 0.995 |

*"Season" column excludes game 01103572's stats, and the "Avg" in "Season League Avg" refers to the mean.*




## 3. Evaluation

*$GCR$ and $GCR+$ of 0 the catcher put up this game is exactly the single-game league median, meaning it was league average.*

**What went well**

- *Managed to get a call overturned.*
- *The one unsuccessful challenge was only off by 0.559in, making it a very good attempt.*

**What to work on**

- *Missed 2 **challenge-worthy** called balls (should have been challenged).*
- *Considering all other games, the catcher tends to miss **challenge-worthy** called balls inside (for a RHB).*

**Bottom line:** *With regards to ABS, the catcher's performance was overall average.*

**Caveats:** *The overturned call arguably should have been flagged as **challenge-worthy** by our model, which would make his $GCR$ 0.333. This is a matter of adjusting the model's **challenge-worthy** confidence threshold and increasing model precision.*